

MID TERM EXAMINATION

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 35 | Date: 12-03-2026

Total Questions: 10 | Answer all questions.

Section 1: MCQ (5 Questions - 10 Marks)

Q1. [2 marks] (Easy)

Which of the following is an example of a linear data structure?

Options:

A) Tree B) Graph C) Stack D) Hash Table

Q2. [2 marks] (Easy)

The process of inserting an element into a full stack is known as what condition?

Options:

A) Underflow B) Overflow C) Null D) Empty

Q4. [2 marks] (Medium)

What is the primary advantage of using a linked list over an array for dynamic data storage?

Options:

A) Faster random access B) Less memory consumption C) Dynamic size and easier insertions/deletions D) Cache locality

Q7. [2 marks] (Medium)

In a singly linked list, which of the following operations has a time complexity of $O(1)$ if the head pointer is known?

Options:

A) Inserting a node at the end B) Deleting a node from the middle C) Inserting a node at the beginning D) Finding a node by value

Q10. [2 marks] (Hard)

Consider an empty queue implemented using an array of size N . If ' N ' enqueue operations are performed, followed by ' N ' dequeue operations, what is the final position of the front and rear pointers (assuming 0-indexed, front points to the first element, rear points to the last element)?

Options:

A) front=0, rear= $N-1$ B) front= N , rear= $N-1$ C) front= N , rear= N D) front=0, rear=-1

Section 2: Short Answer (5 Questions - 25 Marks)

Q3. [5 marks] (Easy)

Briefly explain the concept of an Abstract Data Type (ADT) with an example.

Q5. [5 marks] (Medium)

Differentiate between a Stack and a Queue based on their data access principles.

Q6. [5 marks] (Medium)

Explain how a circular queue can overcome the "memory wasted" problem of a linear array-based queue when elements are frequently enqueued and dequeued.

Q8. [5 marks] (Hard)

Describe the memory implications and benefits of using a doubly linked list compared to a singly linked list. When would a doubly linked list be preferred?

Q9. [5 marks] (Hard)

Design a pseudo-code function to reverse a singly linked list iteratively.

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